

CHAPTER 13

Table 13.S In many of the problems numerical values of n_i or n_i^2 at 300 K are needed -- see (22b). We give these values here, for the semiconductors in Table 13.1:

Semiconductor	n_i cm^{-3}	n_i^2 cm^{-6}
Si	4.58×10^9	2.10×10^{19}
Ge	1.70×10^{13}	2.89×10^{26}
GaAs	2.56×10^6	6.55×10^{12}
InP	8.41×10^6	7.08×10^{13}
InSb	1.64×10^{16}	2.70×10^{32}

13.1) WEAKLY DOPED SEMICONDUCTOR

When $\Delta n \ll n_i$, the square roots in (27a) and (27b) may be expanded as follows:

$$\begin{aligned}
 [(\Delta n)^2 + 4n_i^2]^{\frac{1}{2}} &= 2n_i[1 + (\Delta n/2n_i)^2]^{\frac{1}{2}} \\
 &\cong 2n_i[1 + \frac{1}{2}(\Delta n/2n_i)^2] = 2n_i + (\Delta n)^2/4n_i \quad .
 \end{aligned}$$

Inserted into (27a) and (27b):

$$\begin{aligned}
 n_e &\cong n_i + \frac{1}{2}\Delta n + (\Delta n)^2/8n_i \cong n_i + \frac{1}{2}\Delta n \\
 n_h &\cong n_i - \frac{1}{2}\Delta n + (\Delta n)^2/8n_i \cong n_i - \frac{1}{2}\Delta n
 \end{aligned}$$

13.2) INTRINSIC CONDUCTIVITY AND MINIMUM CONDUCTIVITY

(a) Replacing n_h in (99) by n_i^2/n_e , from (22c):

$$\sigma = e(n_e \tilde{\mu}_e + n_i^2 \tilde{\mu}_h / n_e) \quad .$$

This has a minimum when

$$\begin{aligned} d\sigma/dn_e &= 0 = e(\tilde{\mu}_e - n_i^2 \tilde{\mu}_h / n_e^2) \quad , \\ n_e &= (\tilde{\mu}_h / \tilde{\mu}_e)^{1/2} n_i \quad , \quad n_h = n_i^2 / n_e = (\tilde{\mu}_e / \tilde{\mu}_h)^{1/2} n_i \quad . \end{aligned}$$

Hence

$$\Delta n = \frac{\tilde{\mu}_h - \tilde{\mu}_e}{(\tilde{\mu}_e \tilde{\mu}_h)^{1/2}} n_i \quad ,$$

$$\sigma_{\min} = 2e(\tilde{\mu}_e \tilde{\mu}_h)^{1/2} n_i \quad .$$

(b) This is lower than the intrinsic conductivity $\sigma_i = e(\tilde{\mu}_e + \tilde{\mu}_h)n_i$ by the factor

$$\sigma_{\min}/\sigma_i = 2(\tilde{\mu}_e \tilde{\mu}_h)^{1/2} / (\tilde{\mu}_e + \tilde{\mu}_h) \quad .$$

(c) We take n_i from Table 13.S:

$$\begin{aligned} \underline{\text{Si}}: \quad \Delta n &= -4.95 \times 10^9 \text{ cm}^{-3} \quad (\text{p-type}), \\ \sigma_{\min} &= 1.18 \times 10^{-6} (\Omega \text{ cm})^{-1}, \quad \sigma_i = 1.34 \times 10^{-6} (\Omega \text{ cm})^{-1}, \\ \sigma_{\min}/\sigma_i &= 0.88 \quad . \end{aligned}$$

$$\begin{aligned} \underline{\text{InSb}}: \quad \Delta n &= -1.65 \times 10^{17} \text{ cm}^{-3} \quad (\text{p-type}), \\ \sigma_{\min} &= 40 (\Omega \text{ cm})^{-1}, \quad \sigma_i = 205 (\Omega \text{ cm})^{-1}, \\ \sigma_{\min}/\sigma_i &= 0.195 \quad . \end{aligned}$$

13.3) RESISTIVITY AND IMPURITY CONCENTRATION

Replacing n_h in (99) by n_i^2/n_e , from (22c):

$$\sigma = 1/\rho = e(n_e \tilde{\mu}_e + n_i^2 \tilde{\mu}_h / n_e) \quad .$$

This is a quadratic equation for n_e , which may be written $n_e^2 - 2an_e + b^2 = 0$, where

$$\begin{aligned} a &= 1/2e\tilde{\mu}_e\rho = 4.00 \times 10^{13} \text{ cm}^{-3} \quad ; \\ b^2 &= n_i^2(\tilde{\mu}_h/\tilde{\mu}_e) = 1.41 \times 10^{26} \text{ cm}^{-6} \quad ; \\ (a^2 - b^2)^{1/2} &= 3.82 \times 10^{13} \text{ cm}^{-3} \quad . \end{aligned} \tag{S1}$$

With these, the two roots of the quadratic equation are

$$n_e = a \pm (a^2 - b^2)^{\frac{1}{2}} = \begin{cases} 7.82 \times 10^{13} \text{ cm}^{-3} , \\ 1.80 \times 10^{12} \text{ cm}^{-3} . \end{cases}$$

The corresponding hole concentrations follow by insertion into (22c):

$$n_h = n_i^2 / n_e = \begin{cases} 3.69 \times 10^{12} \text{ cm}^{-3} , \\ 1.61 \times 10^{14} \text{ cm}^{-3} , \end{cases}$$

which leads to the net impurity concentrations

$$\Delta n = n_e - n_h = \begin{cases} 7.45 \times 10^{13} \text{ cm}^{-3} \text{ (n-type)} , \\ -1.59 \times 10^{14} \text{ cm}^{-3} \text{ (p-type)} . \end{cases}$$

13.4) MASS ACTION LAW FOR HIGH CONCENTRATIONS

If we insert $r = n_e / n_c$ and $\eta = (\mu - \varepsilon_c) / \tau$ into (38), the result may be re-arranged to read

$$\log(n_e / n_c) \cong (\mu - \varepsilon_c) / \tau - n_e / \sqrt{8} n_c + \dots ,$$

$$n_e = n_c \exp[(\mu - \varepsilon_c) / \tau - n_e / \sqrt{8} n_c + \dots] . \quad (S1)$$

On the other hand, when $n_e \geq n_c$, we will certainly have $n_h \ll n_v$; hence (21) remains valid. Multiplying (S1) by (22) gives (39).

Comment. At very high carrier concentrations the energy gap of semiconductors shrinks slightly. It has been found useful in this case to define the quantity represented by the symbol n_i by (22b), even though the value defined in this way is no longer the carrier concentration of the pure semiconductor, but is larger. Instead, it is the carrier concentration that the pure semiconductor would have, if it had the same energy gap as the impure semiconductor. Experience shows that, in GaAs at least, the factor n_i^2 thus defined increases with increasing electron

concentration somewhat more rapidly than the exponential factor in (39) decreases, leading to an overall increase of $n_e n_h$ rather than a decrease (H.C. Casey; personal communication).

13.5) ELECTRON AND HOLE CONCENTRATIONS IN InSb

We have to satisfy the two simultaneous equations (25) and (39):

$$n_e - n_h = n_d^+ = n_c \quad , \quad n_e n_h = n_i^2 \exp(-n_e/\sqrt{8}n_c) \quad .$$

Elimination of n_h and re-arrangement

$$r(r-1) = (n_i/n_c)^2 \exp(-r/\sqrt{8}) \quad , \quad \text{where } r = n_e/n_c. \quad (S1)$$

Numerically, from Tables 13.1 and 13.S: $(n_i/n_c)^2 = (1.64/4.6)^2 = 0.127$. The transcendental equation (S1) may be solved graphically by plotting both sides as functions of r and finding the intersection of the two curves. We proceed here by numerical iteration. To this end we re-arrange (S1) into the form

$$r = f(r) \quad ,$$

and solve this equation by the iterative sequence

$$r_{n+1} = f(r_n) \quad ; \quad n = 0, 1, 2, \dots \text{ etc. } \quad ,$$

starting with a suitable initial value of r_0 . It is not difficult to show that this process converges if $|f'(r)| < 1$ in a sufficiently wide range around the actual solution. A little contemplation shows that the root of (S1) must satisfy $r > 1$, and that for $r > 1$ the condition $|f'(r)| < 1$ is satisfied if we re-arrange (S1) by simply dividing by r and adding 1 to both sides:

$$r = f(r) = 1 + (n_i/n_c)^2 \exp(-r/\sqrt{8})/r \quad .$$

Starting with the ad-hoc value $r = 1$ leads to the rapidly converging sequence

$$r = 1.0891, 1.0793, 1.0803, 1.0802, 1.0802, \text{ etc.}$$

After 8 steps, the iteration would have converged, within 10 significant figures, to 1.080244307. If we simply take $r = 1.08$, we obtain $n_e = 1.08n_c = 4.968 \times 10^{16} \text{ cm}^{-3}$. The corresponding value for n_h is $n_h = (4.968 - 4.6) \times 10^{16} \text{ cm}^{-3} = 3.68 \times 10^{15} \text{ cm}^{-3}$. The Fermi level position is obtained by inserting $r = 1.08$ into (38): $n = \log 1.08 + 1.08/\sqrt{8} = 0.46$, or $\mu = \epsilon_c + 0.46\tau$.

13.6) INCOMPLETE IONIZATION OF DEEP IMPURITIES

The condition

$$\Delta\epsilon_d - \tau \log(n_c/8n_d) \gg \tau$$

implies that, in (52), with the help of (49),

$$8n_d/n_e^* = (8n_d/n_c) \exp(\Delta\epsilon_d/\tau) \gg 1.$$

We may then neglect the two one's in (52) and write

$$n_e/n_d \cong \frac{1}{4}(n_e^*/n_d)(8n_d/n_e^*)^{\frac{1}{2}} = (n_c/2n_d)^{\frac{1}{2}} \exp(-\Delta\epsilon_d/2\tau). \quad (S1)$$

This is the fraction of ionized donors.

Note the occurrence of $\Delta\epsilon/2\tau$ rather than $\Delta\epsilon/\tau$ in what looks like a Boltzmann factor in (S1). It occurs because for deep impurities the Fermi level will be located approximately halfway between the conduction band and the donor level, just as for a pure semiconductor the Fermi level is located approximately halfway between the conduction band and the valence band.

NOTE: In the first three printings of the text, Problem 13.6 contained two parts, (a) and (b). In part (a), $\tau \log(8n_d/n_c)$ should be corrected to $\tau \log(n_c/8n_d)$. Part (b) should be deleted altogether. The semiconductor band model is not a suitable framework for the discussion of impurities in true isolators. The above solution is for the corrected part (a) only.

13.7) BUILT-IN FIELD FOR EXPONENTIAL DOPING PROFILE

A non-uniform doping implies non-uniform electron and hole concentrations, and hence position-dependent conduction and valence band energies ε_c and ε_v , which in turn imply an electric field

$$E = -d\phi/dx = + \frac{1}{e} d\varepsilon_c/dx = + \frac{1}{e} d\varepsilon_v/dx \quad .$$

Here ϕ is the electrostatic potential, and ε_c and ε_v vary with varying ϕ according to $\varepsilon = -e\phi + \text{constant}$. In a p-type semiconductor, when $n_i \ll n_h \ll n_v$, we can obtain ε_v from (21):

$$\varepsilon_v = \mu + \tau \log(n_h/n_v) \quad .$$

Because in thermal equilibrium the Fermi level must be position-independent:

$$E = \frac{1}{e} \frac{d\varepsilon_v}{dx} = \frac{\tau}{e} \frac{d}{dx} \log n_h = \frac{\tau}{e} \frac{d}{dx} \log n_a^- \quad . \quad (S1)$$

Here the last equality assumes $n_h = n_a^-$, that is, negligible donor and electron concentrations, and electrical neutrality.

For an exponential acceptor distribution, $\log n_a^-$ must be a linear function of position, which may be written

$$\log n_a^- = \log n_1 + \frac{x-x_1}{x_2-x_1} \log(n_2/n_1) \quad . \quad (S2)$$

This clearly satisfies the conditions $n_a^-(x_1) = n_1$; $n_a^-(x_2) = n_2$. If (S2) is inserted into (S1),

$$E = \frac{\tau}{e} \frac{\log(n_2/n_1)}{x_2-x_1} = 0.0258 \text{ V} \frac{\log 10^3}{10^{-5} \text{ cm}} = 1.78 \times 10^4 \text{ V/cm} \quad ,$$

a very substantial electric field. Fields of this order occur in the base regions of many bipolar transistors, without any externally applied bias.

Comment. The crucial assumption is the assumption of electrical neutrality, $n_h = n_a^-$. As the discussion of a p-n junction in the main text shows, this cannot be taken for granted when the doping is non-uniform. A detailed analysis shows tht electrical neutrality can be expected only if the ionized impurity concentration varies sufficiently slowly with position that everywhere $dn_e/dx \ll n_e/L$, where L is the Debye length, discussed in the Supplementary Material to this Chapter.

13.8) EINSTEIN RELATION FOR HIGH ELECTRON CONCENTRATIONS

We write the Joyce-Dixon approximation (38) in the form

$$\mu_c = \varepsilon_c + \tau [\log(n_e/n_c) + \sum_{v=1}^{\infty} A_v (n_e/n_c)^v] \quad . \quad (S1)$$

The coefficients A_v were given in the footnote on p.366 of the main text. From (S1),

$$\begin{aligned} \text{grad } \mu_c &= \text{grad } \varepsilon_c + \tau \left[\frac{1}{n_e} + \frac{1}{n_c} \sum v A_v (n_e/n_c)^{v-1} \right] \text{grad } n_c \\ &= \text{grad } \varepsilon_c + (\tau/n_e) \sum_{v=0}^{\infty} v A_v (n_e/n_c)^v \text{grad } n_e , \end{aligned}$$

where in the second line we extended the sum to $v = 0$ by defining $A_0 = 1$.

If this result is inserted into (78), and if we use again (82), the current density may be written in the form (84), with

$$D_e = (\tilde{\mu}_e \tau / e) \sum_{v=0}^{\infty} A_v (n_e/n_c)^v .$$

13.9) INJECTION LASER

We write (88) in the form

$$\mu_v - \varepsilon_v = \mu_c - \varepsilon_c = \eta \tau ; \quad \eta \equiv (\mu_c - \varepsilon_c) / \tau . \quad (S1)$$

For zero ionized impurities, electrical neutrality requires

$$n_h = n_e = r n_c ; \quad r \equiv n_e / n_c . \quad (S2)$$

Because for GaAs $n_c \ll n_v$, the combination of (S1) and (S2) requires $\eta > 0$, that is, the conduction band quasi Fermi level is inside the conduction band, requiring the use of the Joyce-Dixon approximation (38). If we carry only the first-order correction term:

$$\eta = \log r + r / \sqrt{8} . \quad (S3)$$

The valence band quasi-Fermi level is above the valence band; we treat the holes by the classical approximation (21). With the help of (S1) and (S2), (21) may be written

$$\begin{aligned} r &= n_e / n_c = n_h / n_c = (n_v / n_c) \exp(-\eta) ; \\ \log r &= \log(n_v / n_c) - \eta . \end{aligned} \quad (S4)$$

Eqs. (S3) and (S4) form a pair of simultaneous equations for r and η . They may be solved graphically by superimposing, on a $\log r$ -vs- η plot such as Fig. 13.4, the straight line represented by (S4) on the curve representing (S3). The straight line (S4) is easily plotted: It is simply a line going through the point $r = 1$, $\eta = \log(n_v/n_c) = 3.48$, with a slope exactly opposite to the slope of the classical asymptote in Fig. 13.4. One finds an intersection at $\eta \cong 2.3$, $r \cong 3$.

To obtain more accurate values, a numerical solution of the system (S3) and (S4) is preferable which, with a programmable calculator, is probably faster than hand-plotting in any case. By eliminating η from (S3) and (S4) one readily obtains a transcendental equation for r , which may be written

$$0 = f(r) \equiv \log r + ar - b \quad , \quad (S5)$$

where $a = 1/\sqrt{32}$ and $b = \frac{1}{2} \log(n_v/n_c)$. Eq. (S5) is well-suited to solution by Newton's method. Given an approximate root r of (S5), an improved root is

$$r' = r - f(r)/f'(r)$$

where $f'(r) = 1/r + a$. Clearly $r > 1$. Starting with the ad-hoc value $r = 2$, the iteration converges in three steps: $r = 2 \rightarrow 3.0 \rightarrow 3.22 \rightarrow 3.23$, corresponding to $\eta = 2.31$ and $n_e = n_h = 1.48 \times 10^{18} \text{ cm}^{-3}$.

Comment. In actual injection lasers the assumption of zero impurity concentration is not satisfied, and the instructor may wish to discuss or assign this more complicated case. It is easily shown that in this case the equation whose root must be found is

$$0 = \log \frac{n_e}{n_c} + \log \frac{n_h}{n_v} + \left(\frac{n_e}{n_c} + \frac{n_h}{n_v} \right) \frac{1}{\sqrt{8}} \quad (S6)$$

where either n_e or n_h should be eliminated via the neutrality condition

$$n_e - n_h = n_d - n_a \quad .$$

Note that in (S6) the J.D. correction for both electrons and holes has been included.

13.10) MINORITY CARRIER LIFETIME

We write $n_e = n_{e0} + \delta n$, $n_h = n_{h0} + \delta n$, with $n_{e0} n_{h0} = n_i^2$. Then, from (98):

$$t = \frac{\delta n}{R} = \frac{(n_e^* + n_{e0})t_h + (n_h^* + n_{h0})t_e + (t_e + t_h)\delta n}{n_{e0} + n_{h0} + \delta n} \quad .$$

For sufficiently small values of δn , the two δn -terms may be neglected, leading to the low-level lifetime

$$t_\ell = \frac{(n_e^* + n_{e0})t_h + (n_h^* + n_{h0})t_e}{n_{e0} + n_{h0}} \quad . \quad (S1)$$

When δn becomes large compared to the largest of the other four concentrations, the terms not containing δn may be neglected, leading to the high-level lifetime $t_h = t_e + t_h$, not further dependent on δn . While the low-level lifetime depends both on the doping level (through n_{e0} and n_{h0}) and on the energy of the recombination (through n_e^* and n_h^*), the high-level lifetime depends on neither.

The lifetime is independent of δn when $t_\ell = t_h$:

$$(n_e^* + n_{e0})t_h + (n_h^* + n_{h0})t_e = (n_{e0} + n_{h0})(t_e + t_h) \quad ;$$

$$n_e^* t_h + n_h^* t_e = n_{e0} t_h + n_{h0} t_e \quad .$$

Because $n_{e0}n_{h0} = n_i^2$, this is a quadratic equation for either n_{e0} or n_{h0} . Its two root pairs are easily seen to be

$$(a) \quad n_{e0} = n_e^* \quad , \quad n_{h0} = n_h^* \quad ; \quad (S2)$$

$$(b) \quad n_{e0} = (t_e/t_h)n_h^* \quad , \quad n_{h0} = (t_h/t_e)n_e^* \quad . \quad (S3)$$

The corresponding doping levels are

$$(a) \quad \Delta n = n_{e0} - n_{h0} = n_e^* - n_h^*$$

$$(b) \quad \Delta n = (t_e^2 n_h^* - t_h^2 n_e^*) / t_e t_h \quad .$$

As a rule the recombination energy level is sufficiently far away from the intrinsic Fermi level that either $n_e^* \gg n_h^*$ or $n_h^* \gg n_e^*$. Suppose $n_e^* \gg n_h^*$, corresponding to a recombination level above the intrinsic Fermi level. In this case, (a) corresponds to n-type doping and, unless $(t_e/t_h)^2 > n_e^*/n_h^* \gg 1$, (b) corresponds to p-type doping. The dopings are reversed if $n_h^* \gg n_e^*$.

Discussion. Eq. (S1) forms the backbone of minority carrier lifetime theory for many semiconductors. To keep the discussion within reasonable bounds, we assume $t_e = t_h = t^*$. Then

$$t_\ell = (1 + \frac{n^*}{n_{e0} + n_{h0}})t^*, \quad \text{where } n^* = n_e^* + n_h^* \quad .$$

For a given doping (given $n_{e0} + n_{h0}$) and given t^* , the lifetime depends on the energy of the recombination level, having a minimum when n^* has a minimum, that is, for energies near the intrinsic Fermi level. Such levels have a minimum effect on the electron and hole concentration, but are particularly recombination-active. Conversely, shallow donors and acceptors, as used for intentional doping, have large values of either n_e^* (donors) or n_h^* (acceptors); they are therefore recombination-inactive.

For a given energy (given n^*) and a given t^* , the lifetime depends on doping. It has a maximum when $n_{e0} + n_{h0}$ has a minimum, that is, for an intrinsic semiconductor:

$$t_{\ell i} = (1 + n^*/2n_i)t^* .$$

If the recombination energy level is sufficiently far from the intrinsic Fermi level, we will have $n^* \gg 2n_i$ and hence $t_{\ell i} \gg t^*$. The lifetime then decreases with increasing doping at least until the net doping exceeds the value of n^* . The lifetime levels out at $t = t^*$ for $|\Delta n| \gg n^*$.

According to (S2), and (S3) -- for $t_e = t_h$ and $n^* \gg n_i$ -- the net doping level $|\Delta n| \cong n^*$ is also the doping level for which the lifetime becomes independent of δn . For lower doping levels the lifetime decreases with increasing δn , for higher doping levels it increases.

Because of the temperature dependence of n_e^* or n_h^* , whichever is bigger, the lifetimes are highly temperature-dependent, increasing with increasing temperature, unless t^* has a strong temperature dependence of its own in the opposite direction.

13.11) ELECTRON-HOLE PAIR GENERATION

(a) With $n_e^* = n_h^* (=n_i)$, $t_e = t_h = t$, and $n_e = n_h = 0$, (98) becomes

$$R_s = -n_i/2t .$$

(b) In an n-type semiconductor from which only the holes have been swept out, we have $n_e = n_d^+$, $n_h = 0$, hence

$$R_n = -n_i^2/(2n_i + n_d^+)t ,$$

$$R_s/R_n = 1 + n_d^+/2n_i \rightarrow n_d^+/2n_i ,$$

where the last result holds for $n_d^+ \gg n_i$.

(c) Taking n_i from Table 13.S:

$$R_s/R_n \cong 10^{16}/2 \times 5 \times 10^9 \cong 10^6 \quad . (!)$$

Comment. The effect contained in this problem is essential for the understanding of the current-voltage characteristic of the p-n junctions in many semiconductors, particularly Si.

DEBYE LENGTH

We consider the electrostatic potential near the edges of a p-n junction, where it deviates from the value deep inside a uniformly doped bulk semiconductor by an amount small compared to τ/e . For $e\phi \ll \tau$ the exponential in (65) may be expanded:

$$\exp(e\phi/\tau) = 1 + e\phi/\tau + \frac{1}{2}(e\phi/\tau)^2 + \dots$$

Inserted into (65):

$$\left(\frac{d\phi}{dx}\right)^2 = -\frac{2en_d}{\epsilon} \left[-\frac{\tau}{2e} \left(\frac{e\phi}{\tau}\right)^2 \right] = \frac{e^2 n_d}{\epsilon \tau} \phi^2 ;$$

$$d\phi/dx = \pm \phi/L, \text{ where } L = (\epsilon \tau / e^2 n_d)^{1/2}.$$

This has solutions of the form

$$\phi(x) = \phi(0) \exp(\pm x/L).$$

The quantity L has the dimension of a length. It is called the Debye length, after an analogous quantity that occurs in the Debye-Hückel theory of electrolytes. For $n_d = 10^{16} \text{ cm}^{-3}$, $\epsilon = 10\epsilon_0$, $T = 300 \text{ K}$ we have $L = 378 \text{ \AA}$. In general:

$$L = 378 \text{ \AA} \times \left(\frac{\epsilon}{10\epsilon_0} \times \frac{T}{300\text{K}} \times \frac{10^{16} \text{ cm}^{-3}}{n_d} \right)^{1/2}.$$